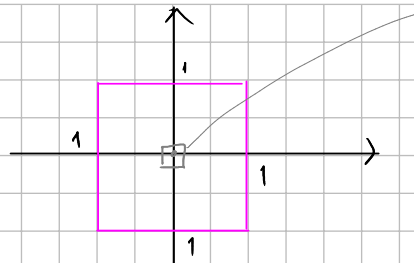


$$u = a(x^2 + y^2)$$

$$v = bxy$$

$$w = 0 \rightarrow$$



$$\underline{u} = \begin{bmatrix} u_x \\ u_y \\ u_z \end{bmatrix} = \begin{bmatrix} u \\ v \\ w \end{bmatrix}$$

Pequeñas deformaciones

$$\epsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$$

$$\epsilon_{xx} = \frac{\partial u_x}{\partial x}$$

$$\epsilon_{xy} = \frac{1}{2} \left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} \right)$$

$$\epsilon_{xx} = 2ax = 2ax$$

$$\epsilon_{xy} = \frac{1}{2} (2ay + by) = \frac{1}{2} (2a + b)y$$

$$\epsilon_{zz} = \frac{\partial u_z}{\partial z} = 0$$

No depende de z

$$\epsilon_{xz} = \frac{1}{2} \left(\frac{\partial u_x}{\partial z} + \frac{\partial u_z}{\partial x} \right) = \frac{1}{2} (0 + 0) = 0$$

$$\epsilon_{xx} \begin{pmatrix} x & y & z \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} = 0$$

$$\therefore = 0$$

$$(\underline{\epsilon} - \epsilon^{(k)} \underline{I}) \cdot \underline{\eta}^{(k)} = 0$$

$$\det(\underline{0} - \epsilon^{(k)} \underline{I}) = 0$$

$$-\epsilon^{(k)3} \det(\underline{I}) = 0$$

$$1$$

$$-\epsilon^{(k)} = 0$$